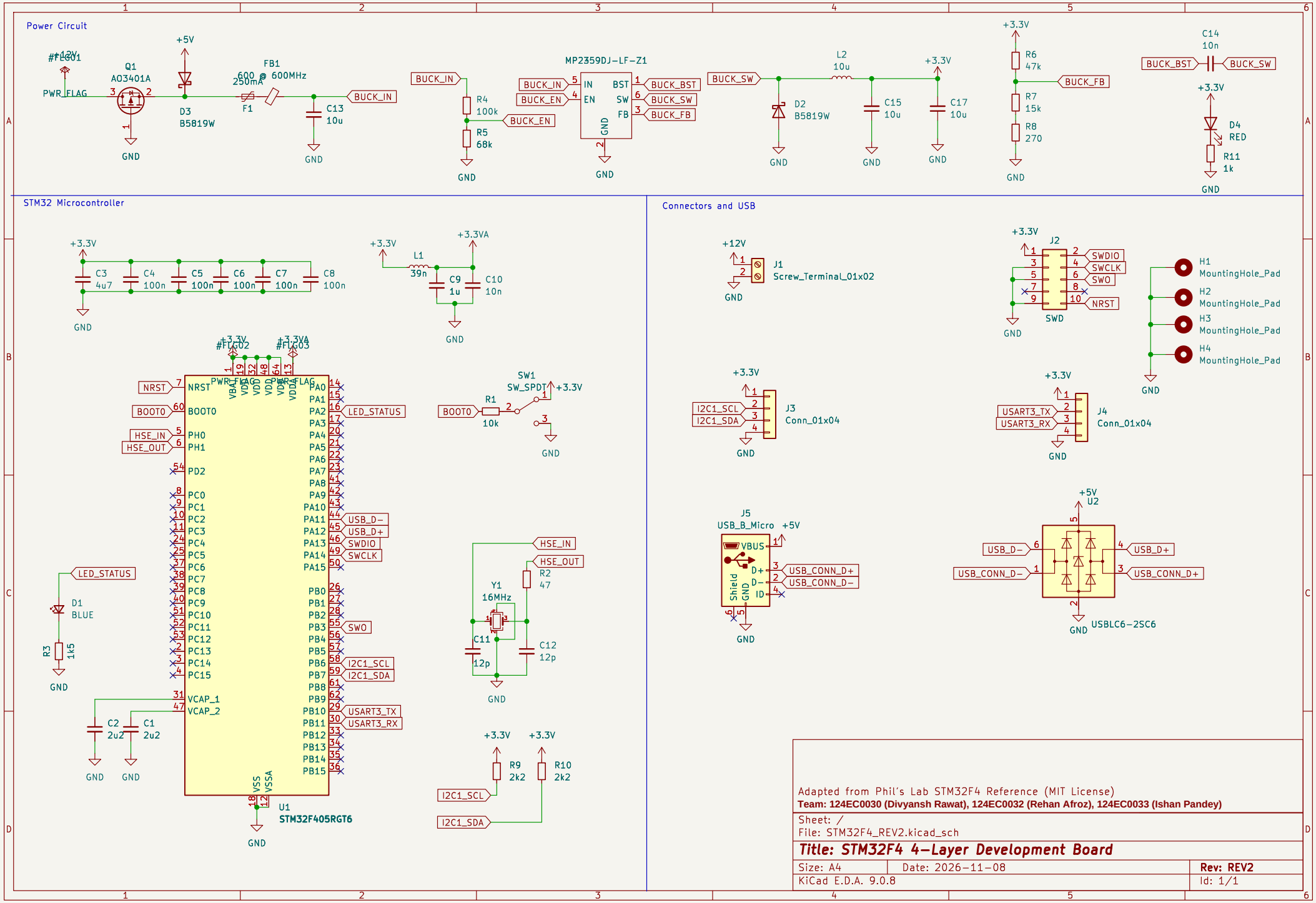

Power Circuit
STM32 Microcontroller
Connectors and USB
Q1
AO3401A
PWR_FLAG
D3
B5819W
FB1
600 @ 600MHz
250mA
F1
C13
10u
BUCK_IN
R4
100k
R5
68k
BUCK_EN
MP2359DJ-LF-Z1
BUCK_BST
BUCK_SW
BUCK_FB
L2
10u
D2
B5819W
C15
10u
C17
10u
R6
47k
R7
15k
R8
270
C14
10n
D4
RED
R11
1k
C3
4u7
C4
100n
C5
100n
C6
100n
C7
100n
C8
100n
L1
39n
C9
1u
C10
10n
U1
STM32F405RGT6
SW1
SW_SPDT
R1
10k
LED_STATUS
D1
BLUE
R3
1k5
Y1
16MHz
C11
12p
C12
12p
R2
47
C2
2u2
C1
2u2
R9
2k2
R10
2k2
I2C1_SCL
I2C1_SDA
USB_D-
USB_D+
SWDIO
SWCLK
SWO
USART3_TX
USART3_RX
J1
Screw_Terminal_01x02
J2
SWD
J3
Conn_01x04
J4
Conn_01x04
J5
USB_B_Micro
U2
USBLC6-2SC6
USB_CONN_D+
USB_CONN_D-
H1
MountingHole_Pad
H2
MountingHole_Pad
H3
MountingHole_Pad
H4
MountingHole_Pad
Adapted from Phil's Lab STM32F4 Reference (MIT License)
Team: 124EC0030 (Divyansh Rawat), 124EC0032 (Rehan Afroz), 124EC0033 (Ishan Pandey)
Sheet: /
File: STM32F4_REV2.kicad_sch
Title: STM32F4 4-Layer Development Board
Size: A4
Date: 2026-11-08
Rev: REV2
KiCad E.D.A. 9.0.8
Id: 1/1